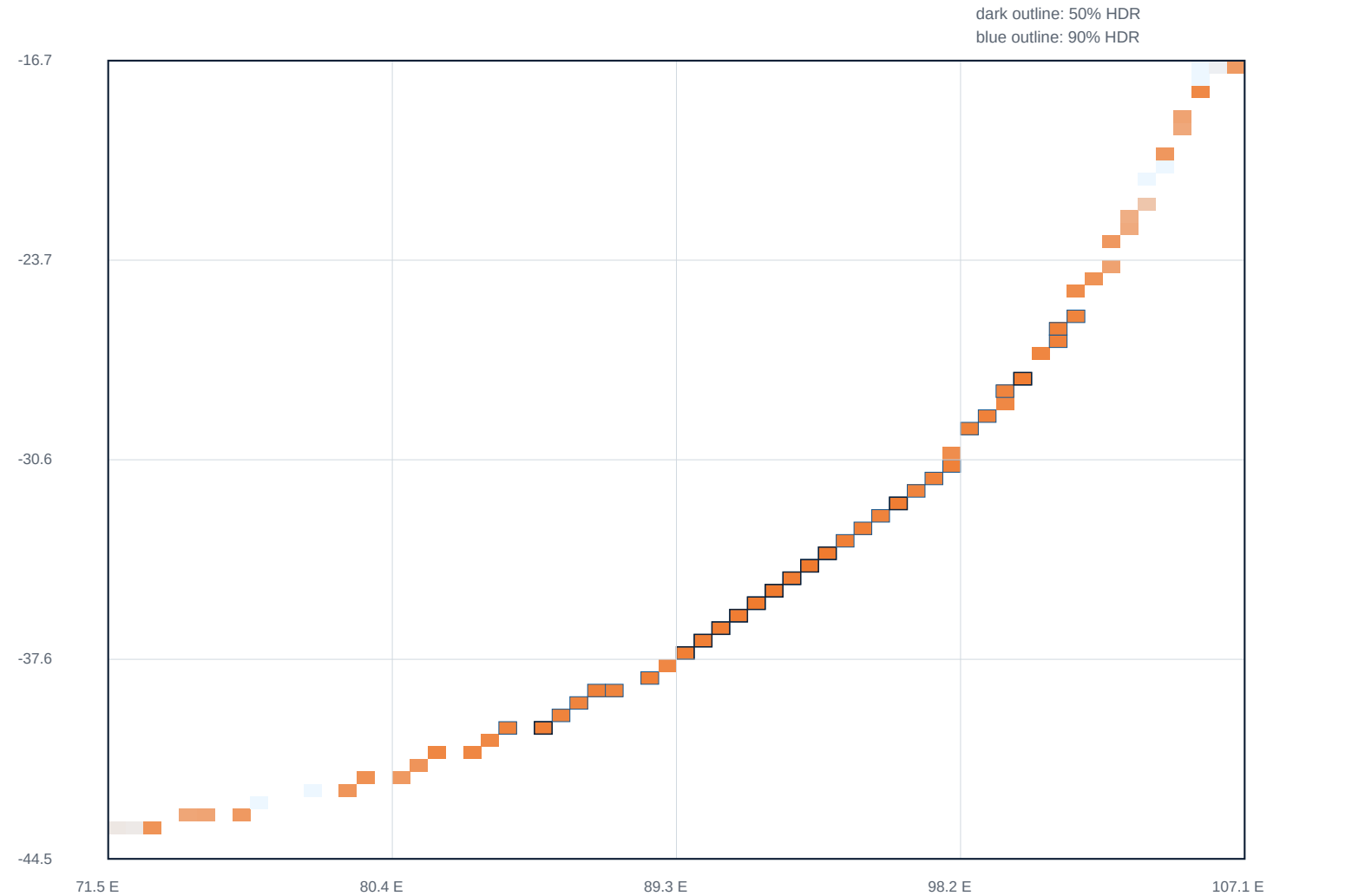


# MH370 flaperon drift diagnostic - native-grid HYCOM

Uncalibrated family-specific diagnostic; currents=hycom\_gofs31\_reanalysis | isotope=enabled as a neutral-or-penalizing screen

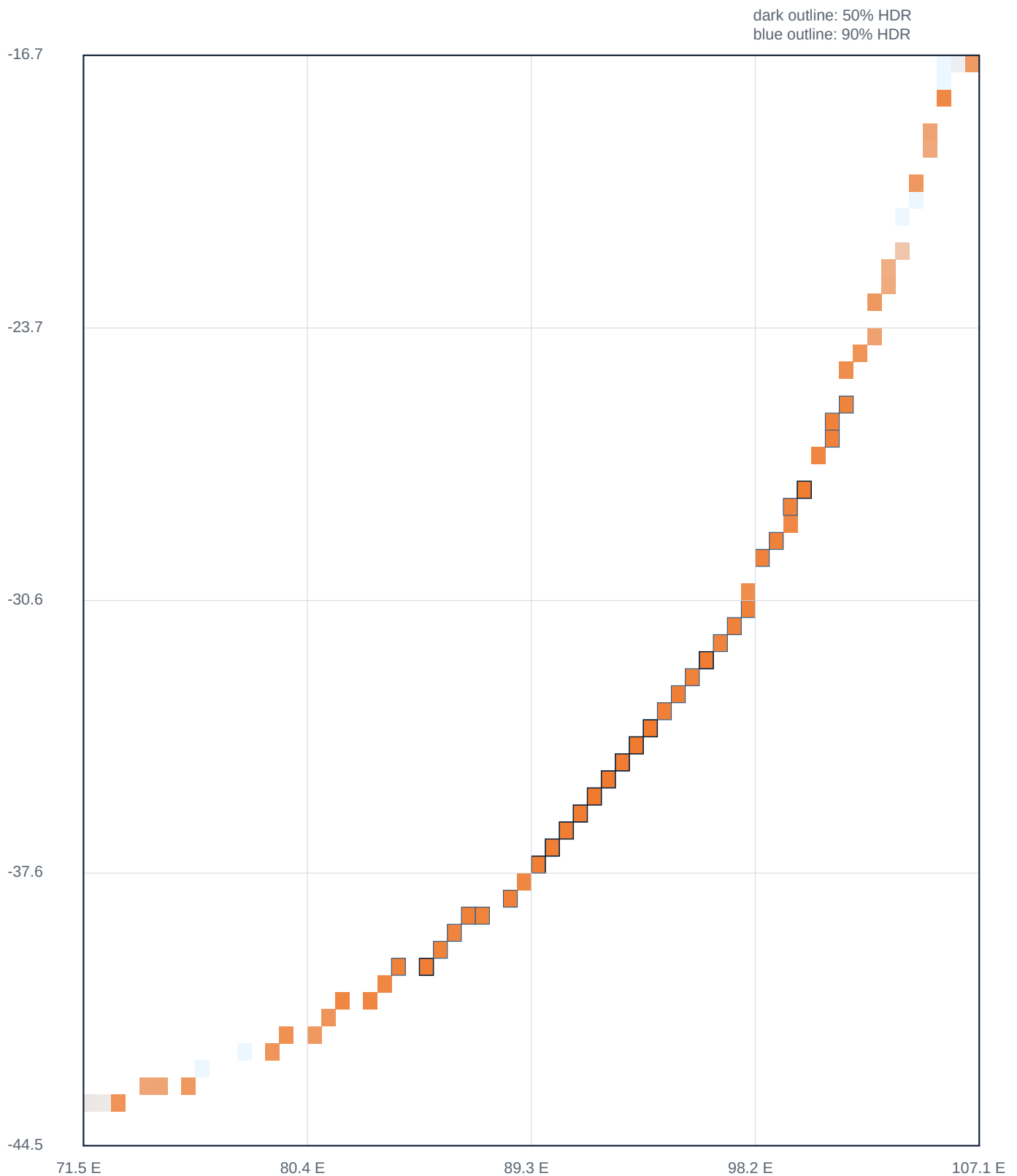
## Estimate at a glance

|                                                     |                                    |
|-----------------------------------------------------|------------------------------------|
| CURRENT FAMILY                                      | PEAK SOURCE CELL                   |
| hycom_gofs31_reanalysis                             | arc-160                            |
| PEAK POSITION                                       | WEIGHTED TRAJECTORIES              |
| -34.463 deg , 93.671 deg                            | 1024 ? 77 cells                    |
| MEAN IMPORTANCE-WEIGHTED FIELD-TERMINATION FRACTION | MEAN PROPOSAL TERMINATION FRACTION |
| 72.4%                                               | 57.4%                              |
| ARRIVAL SAMPLER                                     | PEAK CORRECTION ESS                |
| AdaptiveSmc                                         | 382.6                              |



# Posterior geography

Highest-density regions are computed from the weighted particle population.

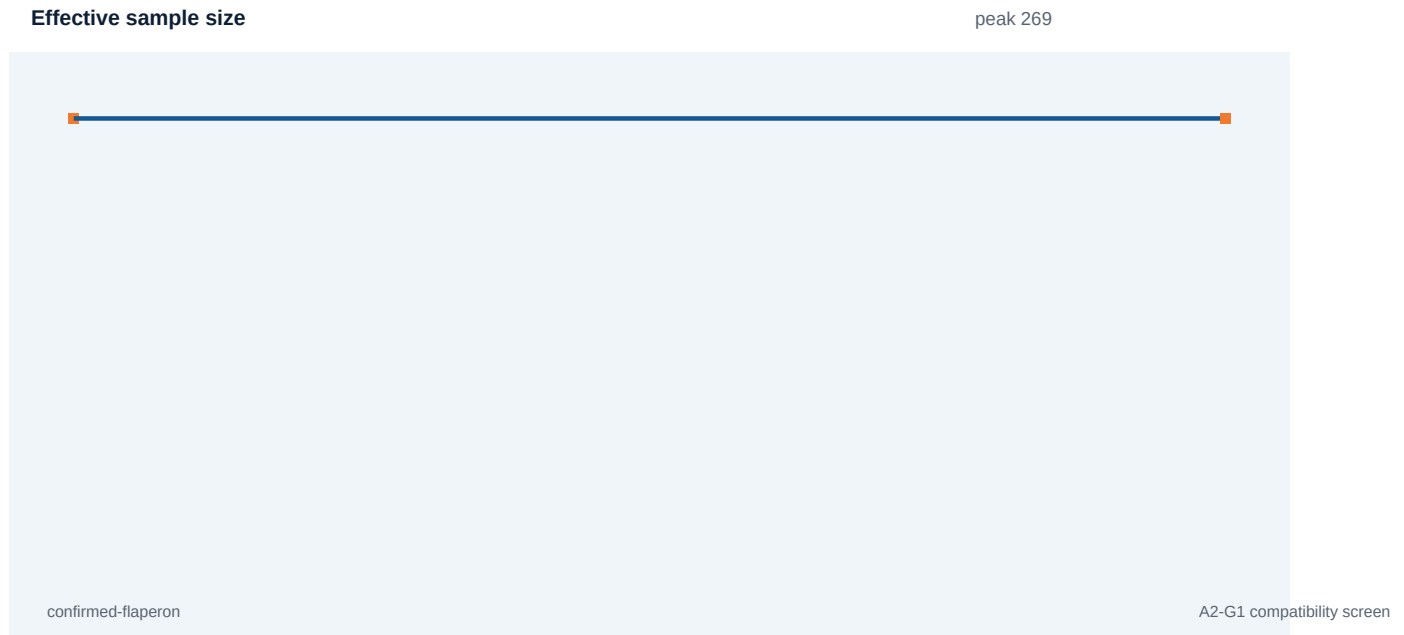


## Interpretation

The map is a probability density over the modelled impact state. It is not a searched-area elimination map, and structural model alternatives must be compared separately.

# Numerical behaviour and conditions

Adaptive likelihood tempering, particle diversity, and declared evidence conditions.



## Evidence conditions

- Current family selected explicitly: hycom\_gofs31\_reanalysis
- Barnacle isotope: enabled as a neutral-or-penalizing screen conditional on represented flaperon arrival
- direct windage disabled
- distinct Stokes velocity field enabled
- explicit land fraction and calibrated beaching field enabled
- Current interpolation across timestamp gaps greater than 36.0 h is outside-time support
- compatible=657 marginal=10 rejected=0 missing\_SST=315
- Isotope sensitivities: seawater SD=0.20?, calibration SD=1.00 deg C, residual SD=1.50 deg C correlated over 14.0 d, SST SD=0.50 deg C, shell smoothing=7.0 d, compatible/rejection simultaneous confidence=0.950/0.995, maximum marginal/rejection penalties=0.693/4.605 log units

## Material limitations

- Diagnostic source-compatibility result, not a calibrated final MH370 posterior.
- Current families are reported separately and are never averaged using subjective weights.
- A2-G1 covers only the final months; the screen is neutral for any compatible chronology and never density-ranks compatible paths.
- No field-unavailable trajectory is relabelled as beaching; termination fractions and reasons remain in the JSON artifact.
- This is every eighth available 3-hour analysis, not a daily mean; exact supplemental analyses repair two stride-created gaps, leaving variable 24-30 hour spacing and phase changes.
- The GOFS 3.1-like reanalysis documents known deep-water and Philippine Sea issues; predictive drifter validation is required before primary use.
- Land remains the packed missing sentinel. No missing ocean value is converted to zero current.
- Stokes drift, direct windage, and coastal beaching are not included in this current field and require distinct explicit inputs.